

PAX path stability — run report

RUN

230655_pax-path-hold_pax-path-hold

Status: completed | Started: 2026-09-23T22:06:55.963226+00:00

PURPOSE

Hold both outputs at zero and record PAX/PD for each manual path condition. Duration is per condition.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Photodiode connected to configured RP input.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

DATA

723 recorded samples; path_a_only: 241, path_b_only: 241, both_paths: 241

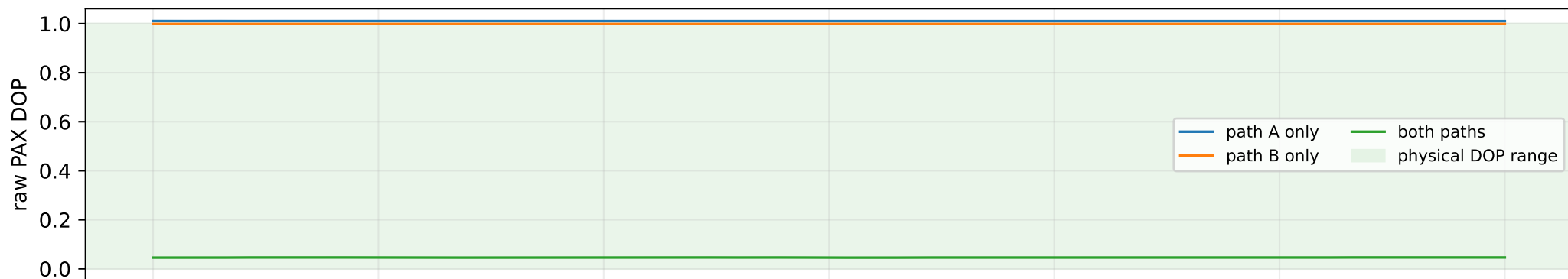
Source: data.csv; full acquisition settings: recipe.json; status: run.json

RAW DOP

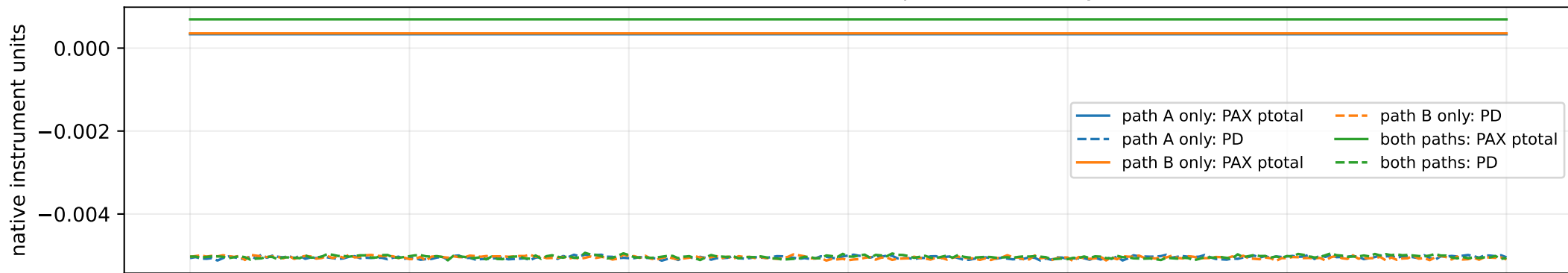
Mean 0.6850; range 0.0444 to 1.0109. 241 samples outside 0-1. Telemetry is not a new calibration.

Fixed-state PAX path diagnostic — 230655_pax-path-hold_pax-path-hold

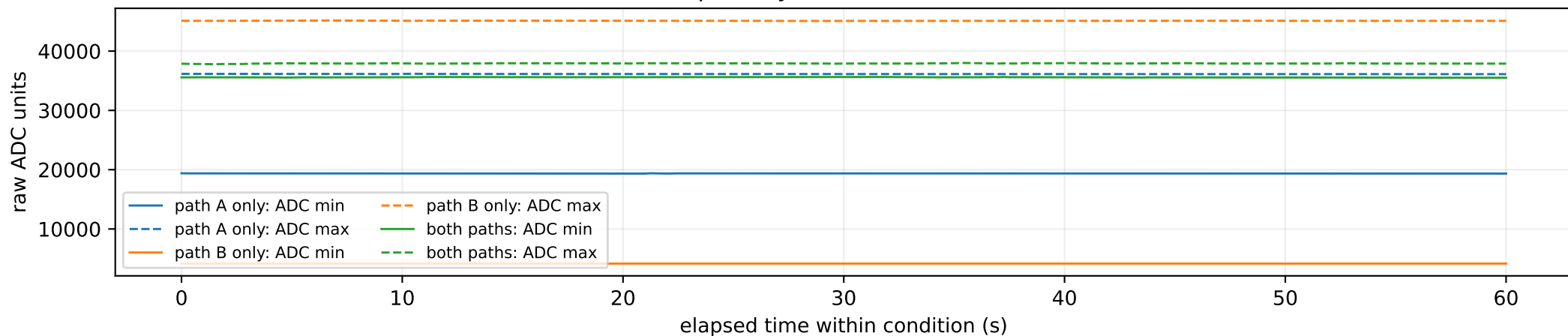
Raw DOP while actuator commands are fixed at 0 V



Concurrent, uncalibrated power telemetry



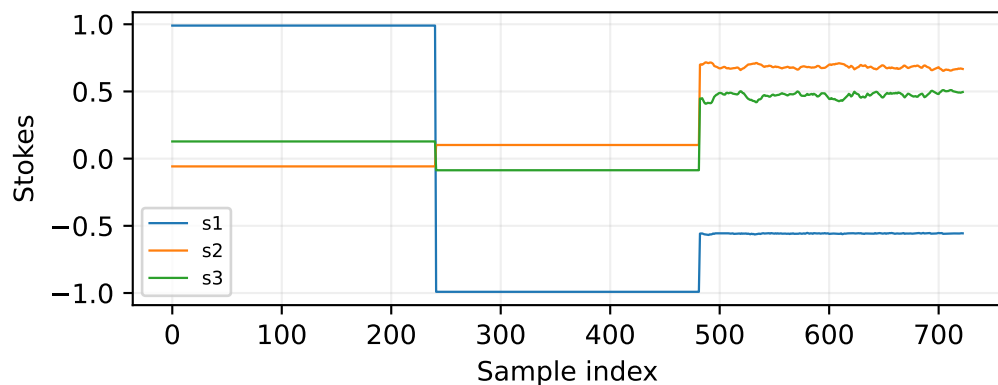
PAX primary-record ADC extrema



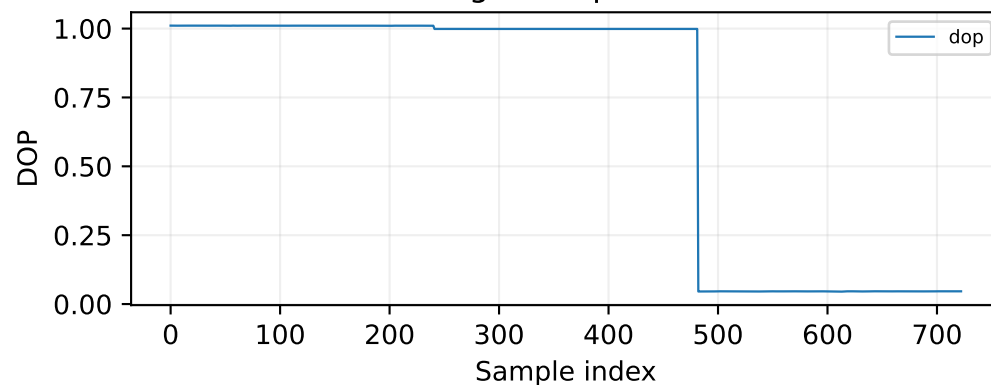
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

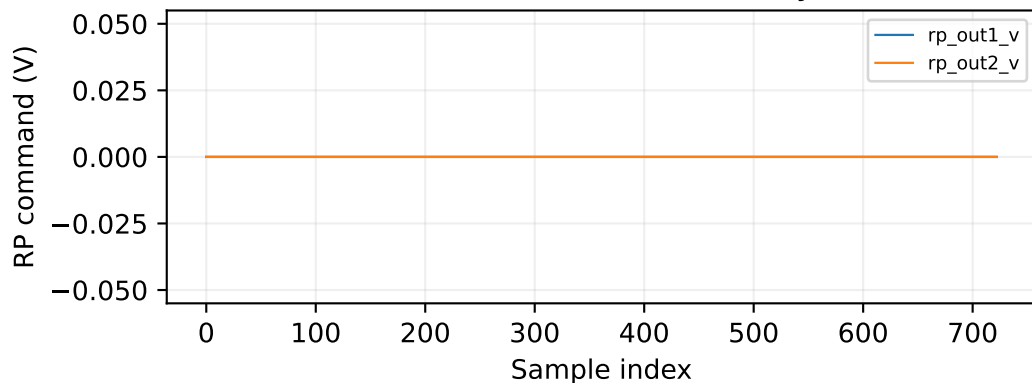
Normalized Stokes



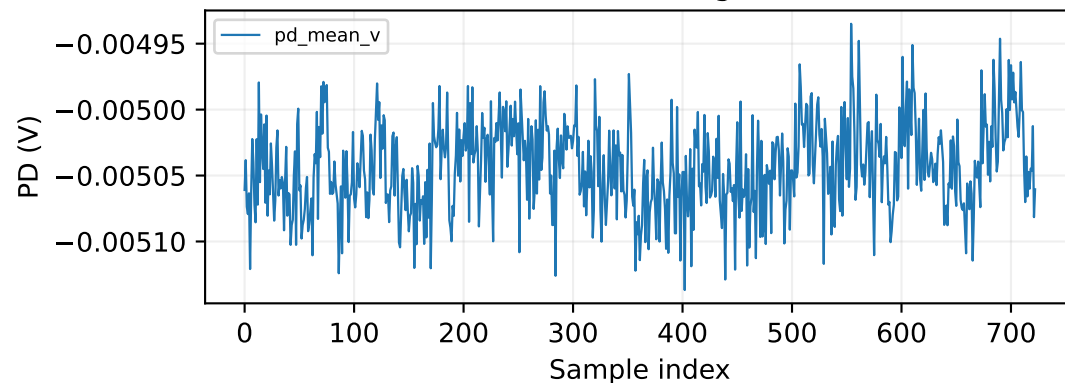
Raw degree of polarization



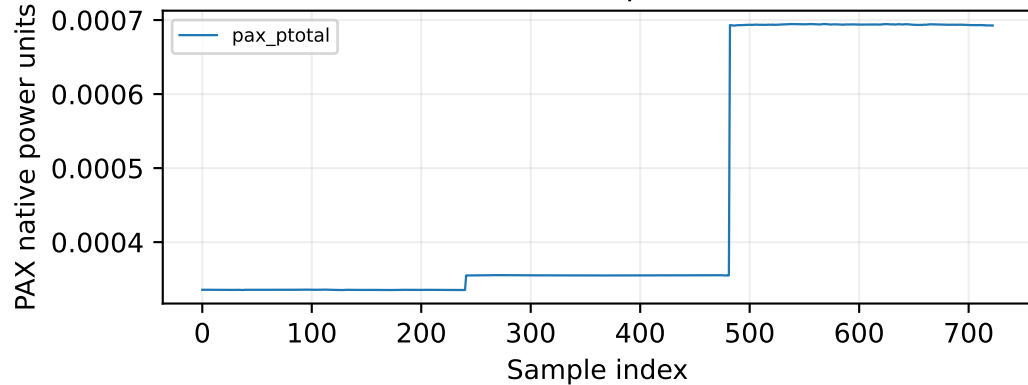
Actuator command history



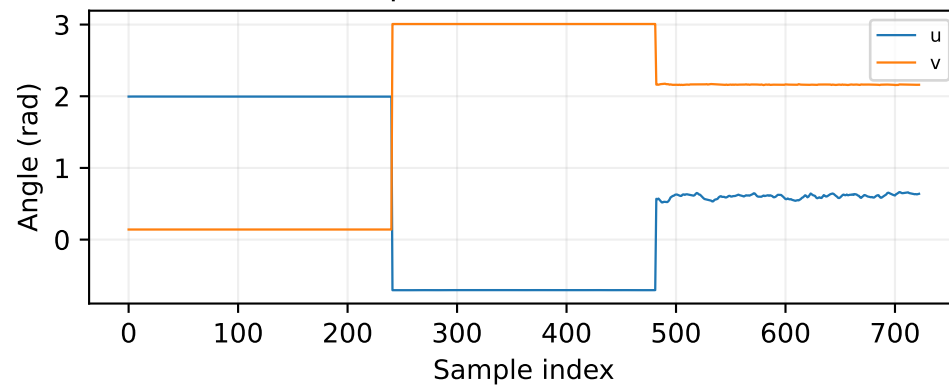
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"pax-path-hold"
duration_s	60.0
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	12.2
phi2_v_lambda	12.2
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
pd_input	"in1"
pd_nd_optical_density	2.0
pd_scope_duration_s	0.01
pd_scope_decimation	64
pd_scope_timeout_s	2.0
pax_path_hold_sample_period_s	0.25